MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 1L4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, RF, sharp cutoff

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	-f,g3,sh	a	g2	nc	-f,g3,sh	g1	+f

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ik
Unit:	Vdc	Vdc	Vdc	Vdc	mAdc
Maximum:	1.6	---	100	120	6.5
Minimum:	1.2	---	---	---	---
TEST CONDITIONS:	1.4	0	90	90	---

GENERAL:

Qualification - Not required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	1	Eb = Ec2 = 110 Vdc; Ec1 = -2.0 Vdc	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (1) (anode)	-		0.65	II	Ib	2.9	6.1	mAdc
1306	Transconductance (2)	1	Ef = 1.1 Vdc; t = 5 (max)	0.65	II	Sm	690	1,290	µmhos
©1246	Audio frequency noise	2	Eca1 (0 vu) = 725 mVac Eca1 (50 mW) = 200 mVac	1.0	II	EB	---	0	vu
1201	Short and discontinuity detection	-	40 G	0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Rp = 10,000 ohms	---	---	Ep	---	500	mVac
1301	Filament current	-		---	---	If	44	56	mAdc
1256	Electrode current (grid)	3	Ec2 = 45 Vdc	---	---	Ic1	---	+0.8	µAdc
1256	Electrode current (screen)	-		---	---	Ic2	1.0	3.0	mAdc
1256	Electrode current (2) (anode)	-	Ec1 = -10 Vdc	---	---	Ib	0	50	µAdc
1256	Electrode current (3) (anode)	-	Ec1 = -6 Vdc	---	---	Ib	100	---	µAdc
1306	Transconductance (1)	-		---	---	Sm	760	1,290	µmhos
1311	Anode resistance	-		---	---	rp	0.17	---	Meg
1331	Direct-interelectrode capacitance	-	Shield No. 316 No shield No shield	---	---	Cgp Cin Cout	--- 2.9 5.3	0.01 4.3 9.7	pF pF pF
1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
1121	Base strain	-		---	---	---	---	---	---
2126	Glass strain	-		4.0	I	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group A; Ef = 1.4 Vdc or Vac	---	---	---	---	---	---
---	Life-test end point: (500 hours)								
1306	Transconductance (1)	-		---	---	Sm	670	---	µmhos

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Ebb = Ecc2 = 135 Vdc; Rp = 1.0 Meg; Rg1 = 1.5 Meg; Rg2 = 4.7 Meg (by-passed with 0.1 µF capacitor to -f). Set amplifier gain for 50 mw output with Esig = 200 mVac.
3. The total resistance of the grid circuit including meter, shall not exceed 10,000 ohms.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - WC ,AR
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Army - EL

Agent:
DLA - ES

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